

Predicting Paddy Yield with Machine Learning: An Integrated Analysis of Climate, Agricultural Management, and Ensemble Models

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Abstract

This study investigates which combinations of environmental conditions and agricultural management practices predict paddy yield to support precision agriculture. Historical farming records were analyzed to identify the factors associated with yield performance and to evaluate predictive approaches for agricultural decision-making. The results indicate that human-controlled management practices strongly driving production. From an agricultural perspective, these findings suggest predictive approaches taken can allow farm managers to transition from reactive farming to proactive, data-driven planning by anticipating harvest volumes. A primary limitation of this research is its reliance on static, historical records without incorporating dynamic macroeconomic indicators such as fluctuating fertilizer or labor costs. Future precision agriculture systems must integrate live financial data into these predictive models to provide decision-support for agricultural stakeholders.

Introduction

Rice is one of the most important staple crops in the world, playing a central role in global food security. Because rice production being closely tied to environmental conditions, changes in climate and weather patterns can create serious challenges for farmers, policymakers, and food systems. Temperature, rainfall, humidity, and wind speed can influence rice growth, affecting both crop development and final yield. Rice yield prediction is important because accurate yield estimates can support better agricultural planning, resource management, and climate adaptation. As climate variability increases, understanding how environmental and

farming-related factors affect paddy production becomes essential for improving productivity. This topic is also significant within the field of precision agriculture, where data-driven methods help farmers make more informed decisions. By examining the factors associated with paddy yield, this study contributes to a better understanding of sustainable rice production and its importance for future food security.

Literature Review

Previous research shows paddy yield is shaped by environmental conditions and management practices acting together. Muthukumaran et al. (2023) emphasizes the role of temperature, rainfall, humidity, and wind alongside fertilizer application, soil type, and seed rate. These factors influence physiological processes such as photosynthesis, water uptake, and nutrient absorption. Jeong et al. (2016), Sah et al. (2024), and Sahoo et al. (2024) further show these influences that operate across regional and broader scales. Their findings reinforce the importance of considering diverse and variable growing conditions when examining yield outcomes. Management practices play an equally critical role in determining yield outcomes.

Fertilizer application affects nutrient availability, soil type determines nutrient composition, and management decisions can enhance or limit crop growth. Sah et al. (2024) and Sahoo et al. (2024) further argue that paddy yield reflects complex and interconnected effects in which environmental and management factors interact rather than operate independently. These interactions highlight the need for integrated approaches that account for both climate variability and management decisions when analyzing paddy yield performance. Building on this foundation, this study addresses how environmental and cultivation-related factors influence paddy productivity under varying agricultural conditions. Specifically, the study examines how climatic conditions during different crop growth stages affect yield; identifies the cultivation and

soil management practices that contribute the most strongly to higher productivity; evaluates the nature of the effect of the agricultural management practices have on paddy yield; compares the effectiveness of different predictive approaches in estimating paddy yield from agricultural field data; and explores how yield prediction can support precision agriculture decision-making.

This study hypothesizes that paddy yield is more accurately explained when both environmental conditions and cultivation practices are considered together rather than separately. The study also hypothesizes that management practices play a key role in paddy productivity (Muthukumaran et al., 2023). In addition, the study expects that the interaction between environmental conditions and cultivation practices creates complex effects on yield performance. This is where favorable management practices may enhance productivity under favorable climatic conditions. Finally, the study hypothesizes that predictive approaches capturing these complex interactions among climate and management variables will provide more accurate estimates of paddy yield than approaches based on isolated factors (Wickramasinghe et al., 2021).

Methods

The study used a dataset from the UCI Machine Learning Repository. The dataset consisted of 2,789 paddy production records and 44 variables related to environmental conditions, cultivation practices, and yield outcomes. The data were collected from agricultural production settings and included multiple growing conditions and management scenarios. Predictors included environmental indicators, cultivation characteristics, fertilizer and nutrient-management practices, and crop production outcomes across four growth phases (D1–30, D31–60, D61–90, D91–120). In addition, a yield-per-hectare variable was derived for clustering.

To address the first research question regarding the influence of climate conditions during crop growth stages, predictive and statistical approaches evaluated relationships between environmental conditions and paddy yield. Multiple linear regression was initially used to estimate relationships between climatic conditions and yield outcomes. To address the second research question concerning feature selection and model efficiency, several feature-selection approaches were used to identify influential environmental and cultivation-related factors associated with yield prediction. These approaches included using Recursive Feature Elimination (RFE), and embedded methods such as Least Absolute Shrinkage and Selection Operator (LASSO) regularization and tree-based importance measures. The performance of reduced-feature models was compared with full-feature models.

To address the third and fourth research questions regarding management practices and predictive performance, multiple predictive approaches were compared. Simple generalized models were used to assess the relationships between management intensity and paddy yield to determine how interconnected practices affect paddy yield. The ensemble learning methods and regression-based approaches were evaluated for their ability to model complex interactions among the features. The model performance was assessed using coefficient of determination (R^2), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE). The data were divided into training and testing subsets using an 80/20 split, and a five-fold cross-validation was applied. Lastly, with the fifth research question, Multiple Linear Regression was used as a baseline predictive approach, while Random Forest served as the primary model. Predicted and observed yield values were categorized into low-, medium-, and high-yield categories, compared for agreement, further examined with K-means clustering, profiled by field characteristics, and summarized into decision-support actions.

Linear regression assumptions of linearity, normality, homoscedasticity, and no multicollinearity will be checked using scatter plots, Q-Q plots, residual plots, the Durbin-Watson test, and Variance Inflation Factor (VIF) analysis. If multicollinearity is detected among the features, Principal Component Analysis (PCA) or LASSO regularization will be applied. If linearity or normality violations persist after transformations, tree-based models were used as alternatives (Wickramasinghe et al., 2021). For K-means clustering, variables will be standardized and checked for outliers, with hierarchical clustering as a fallback if clusters are unstable. All analyses will use Python 3.11.9 (pandas 3.0.1, matplotlib 3.10.6, seaborn 0.13.2, statsmodels 0.14.6, scikit-learn) and R 4.3.2, at a significance level of 0.05 with two-tailed tests.

Result/Discussion

The results show that paddy yield in this study is influenced more by agricultural management practices than by broad climate indicators alone. Fertilizer use, nursery preparation, and seed rate repeatedly were the strongest contributors to yield performance (Figure 8). This suggests that, although climate conditions are important for rice growth, farmers may influence yield outcomes more through management decisions. From a precision agriculture perspective, yield improvement should not rely only on weather monitoring, but also on improving resources application in the field. These findings emphasize the value of management-focused decision support for improving productivity and reducing unnecessary input use.

The climate-focused analysis showed that weather variables alone were not enough to explain paddy yield outcomes in this study. Rainfall, humidity, wind speed, and temperature across different growth stages showed weak direct relationships with yield (Figures 3 and 4). This suggests that the available climate information may be too broad to capture short-term field conditions. Rice can be sensitive to extreme rainfall, drought, or temperature changes during

specific growth periods, but 30-day aggregated measurements may hide these short events (Figure 5b). Therefore, agricultural decision systems should combine climate information with more detailed field-level data, such as daily weather and soil moisture data.

The feature-selection and management-practice results further support the importance of actionable agricultural decisions (Figures 6 and 7). The strongest predictors were mostly related to management inputs rather than environmental factors, which indicates that human-controlled practices are central to improving yield performance. This is useful for farmers and agricultural planners because it suggests that monitoring a smaller number of management indicators may be more practical than collecting many weakly related variables. The management-intensity analysis also showed that the relationship between management practices and yield was not simply linear (Figure 2). Increasing input does not always improve yield proportionally, so a more balanced approach is needed.

The prediction results also demonstrate how yield forecasting can support practical agricultural planning and precision-agriculture decision-making. Accurate yield estimates may help farmers, agricultural agencies, and policymakers make earlier decisions about fertilizer use, labor allocation, food-supply planning, and risk management. Predicted yield outcomes were grouped into different categories to support field management (Table 1). The low- and high-yield categories were more clearly distinguished than the medium-yield group, suggesting that these classifications are most useful as a screening tool rather than as a strict classification system. Higher-yield groups were generally associated with stronger management practices, particularly fertilizer-related and pest-management inputs, while climate indicators showed small differences across productivity groups. Low-yield fields may benefit from closer monitoring and management review, while high-yield fields can serve as reference cases for effective practices.

Several limitations should be considered before applying these findings more broadly. First, the climate variables were aggregated and may not reflect actual field-level weather conditions. Second, some management information was summarized into broader measures, which makes it difficult to identify the exact contribution of each individual practice. Third, the findings may reflect patterns specific to this dataset, so they should be validated in other regions and farming systems. Future research should collect more detailed daily climate records, soil information, and field-level management data. It should also test these yield patterns across different rice-growing regions. Overall, this study suggests that paddy yield prediction is most useful when it supports practical management decisions and helps farmers apply resources more efficiently.

Conclusion

The findings of this research conclusively demonstrate that optimizing human-controlled agricultural management practices is the most critical driver for maximizing paddy yield. By transitioning to advanced predictive forecasting tools, agricultural leaders can accurately anticipate harvest volumes based on complex environmental and managerial interactions. Grouping these anticipated yields into clear performance categories allows farm managers to implement highly targeted interventions across different fields. This strategic foresight enables the differential allocation of critical resources, such as water and fertilizers, to areas where they will be most effective, thereby reducing unnecessary operational costs. Ultimately, integrating comprehensive field data into proactive planning frameworks empowers the agricultural sector to enhance food security, improve economic resilience, and ensure sustainable farming practices in the face of environmental unpredictability.

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Tables and Figures

	Records	Mean observed yield/ha	Mean predicted yield/ha	Mean urea input	Mean pest- control input	Dominant variety	Decision-support interpretation
Predicted_yield_category							
Low-yield	930	5734.74	5735.86	62.95	1392.26	ponmani	Possible low-productivity risk group; prioritize closer monitoring, fertilizer review, and pest-management review.
Medium-yield	929	6012.30	6007.73	106.86	2363.19	ponmani	Moderate productivity group; continue monitoring and identify constraints that may prevent movement into the high-yield group.
High-yield	930	6224.58	6233.10	132.76	2936.13	delux ponni	Favorable productivity group; use as reference cases for favorable variety, management, and field-condition profiles.

Table 1. Decision-Support Summary by Predicted Yield Category

This table summarizes low-, medium-, and high-yield prediction groups and their associated productivity levels, management input patterns, dominant rice varieties, and practical agricultural decision-support interpretations.

Model	R ² Score	RMSE (kg)	Rank
Multiple Linear Regression	0.9891	942.15	4th
Support Vector Regression (SVR)	0.9904	884.14	2nd
Random Forest	0.9903	887.05	3rd
XGBoost (Gradient Boosting)	0.9916	826.33	1st ✓

Table 2. Model Performance Comparison on Paddy Yield Prediction (Test Set)

All four models achieved R² scores above 0.989, indicating that each model explains more than 98.9% of the variance in paddy yield. This exceptional overall performance reflects the high quality and predictive richness of the dataset. However, important differences exist among the models in terms of RMSE, which measures the average prediction error in kilograms.

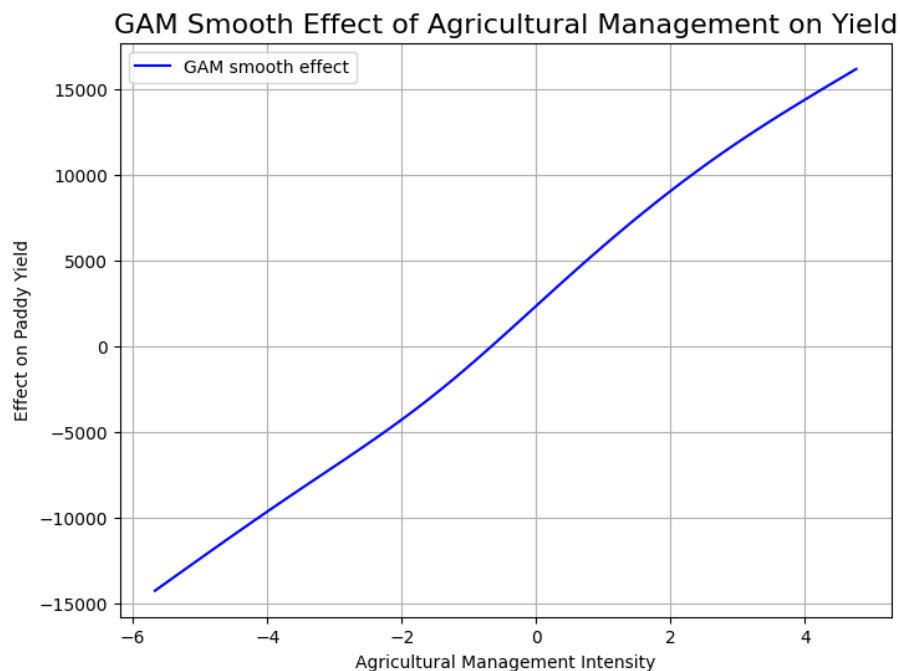


Figure 2. GAM Smooth Effect on Agricultural Management on Paddy Yield

The smooth term of the GAM illustrates the relationship between the practices and yield, showing a generally positive but non-linear trend, with slight curvatures in the line.

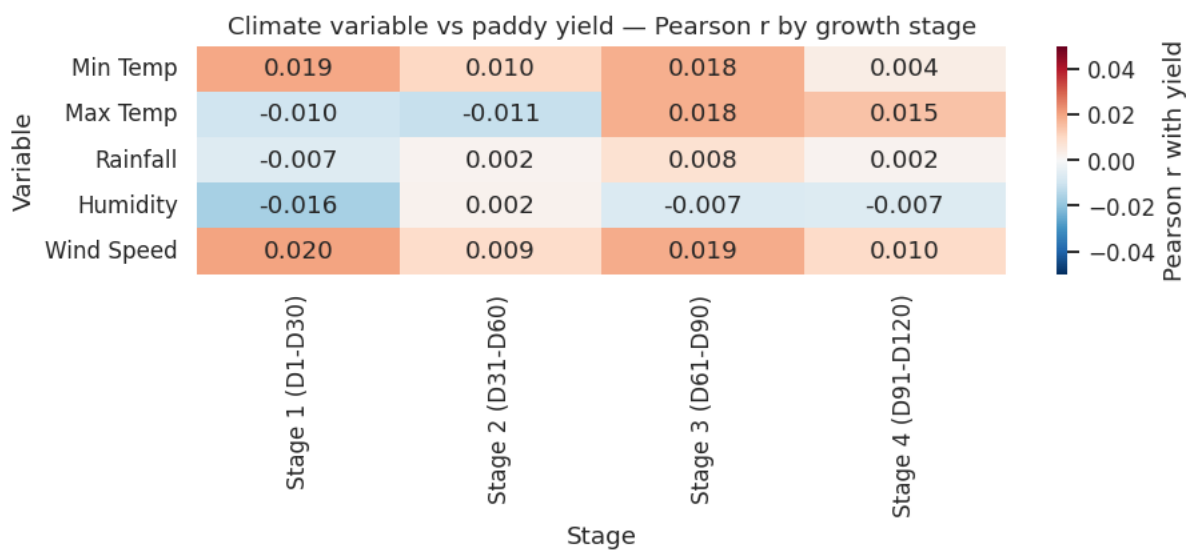


Figure 3. Pearson Correlation Between Each Climate Variable and Paddy Yield, by Growth Stage

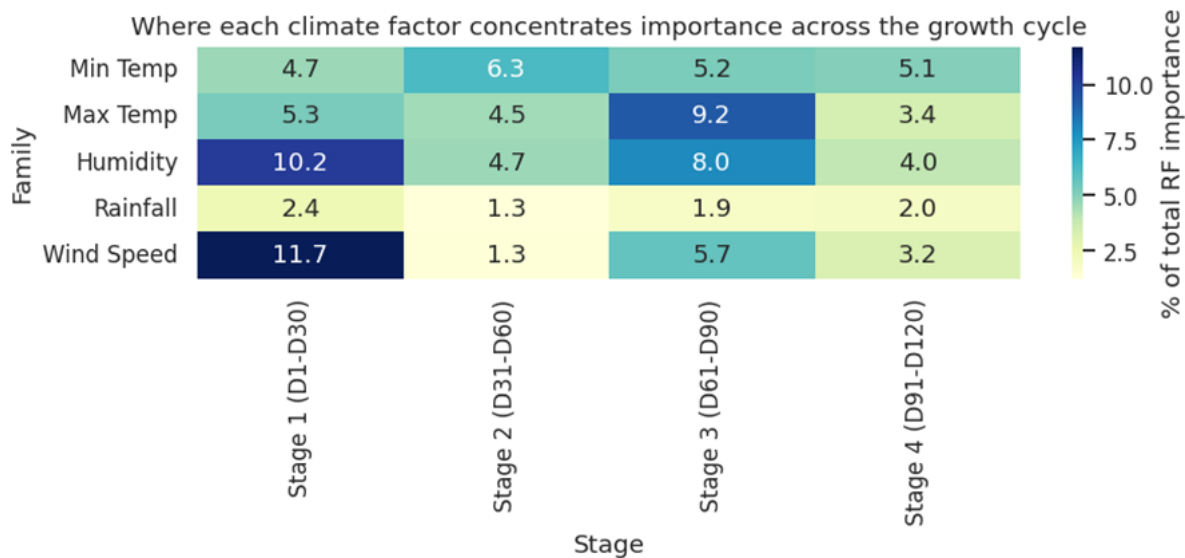


Figure 4. Random Forest Feature Importance by Climate Family and Growth Stage

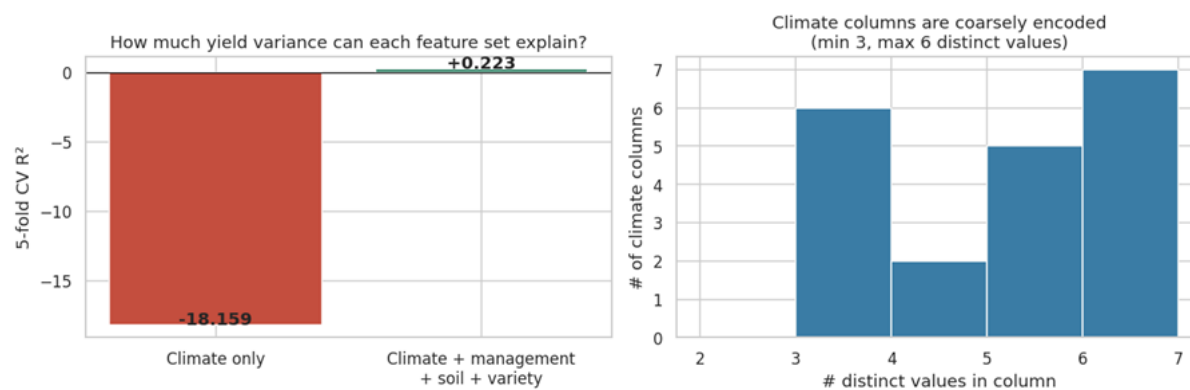


Figure 5. Climate-Only vs. Full-Feature Predictive Performance and Climate-Column Granularity

Supplementary Appendix

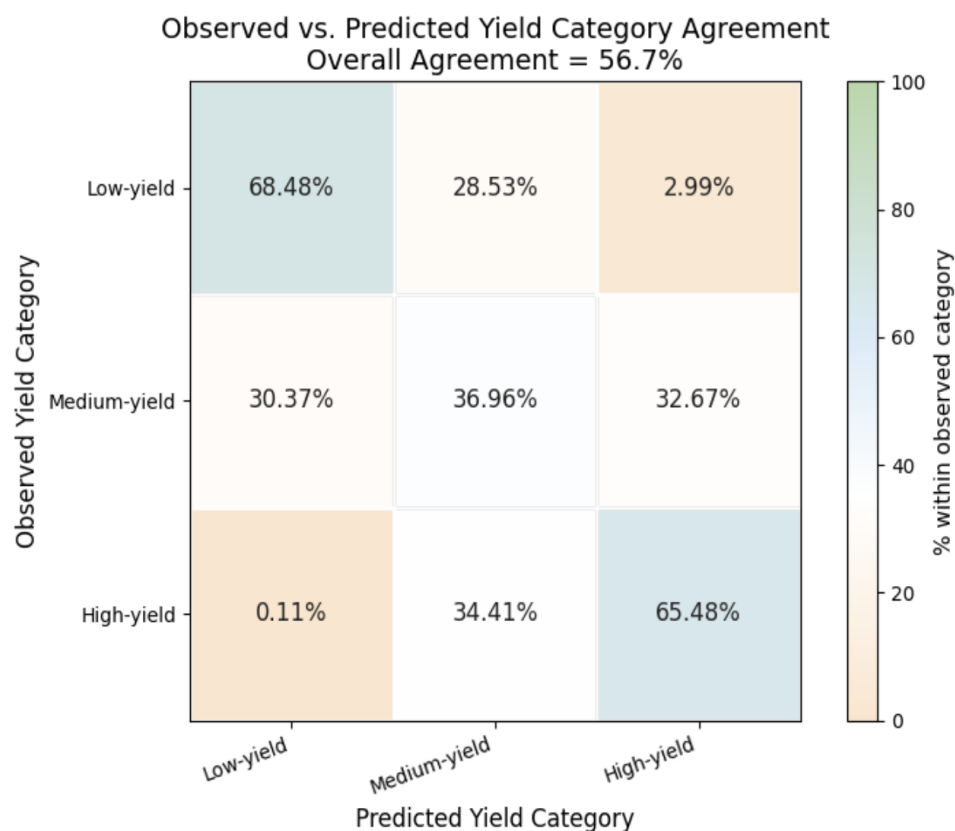


Figure 1. Observed vs. Predicted Yield Category Agreement

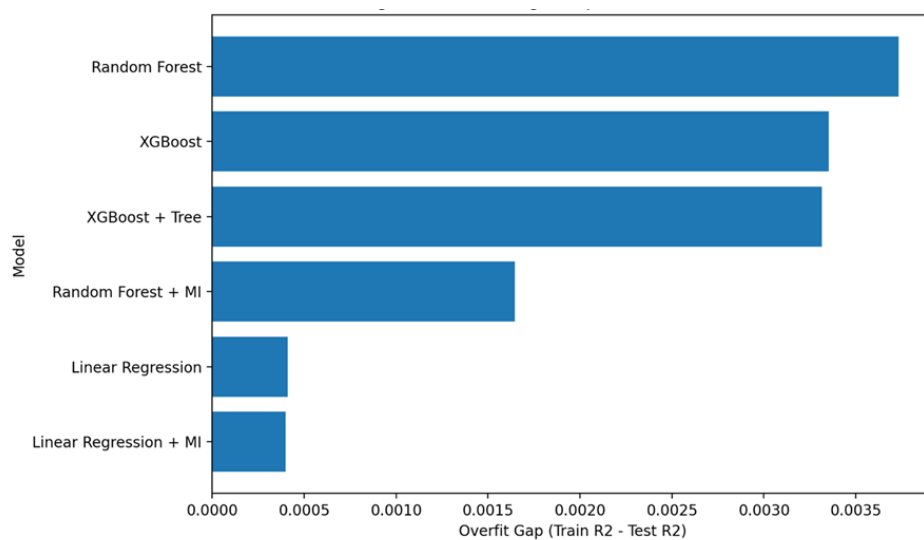


Figure 7. Overfitting Comparison Across Models

Comparison of overfitting across models measured by the gap between training and testing R^2 . Smaller gaps indicate better generalization and less overfitting. The Random Forest model with Mutual Information feature selection showed a smaller overfitting gap than the full-feature Random Forest model, suggesting that feature selection helped reduce model complexity and improve robustness on unseen data.

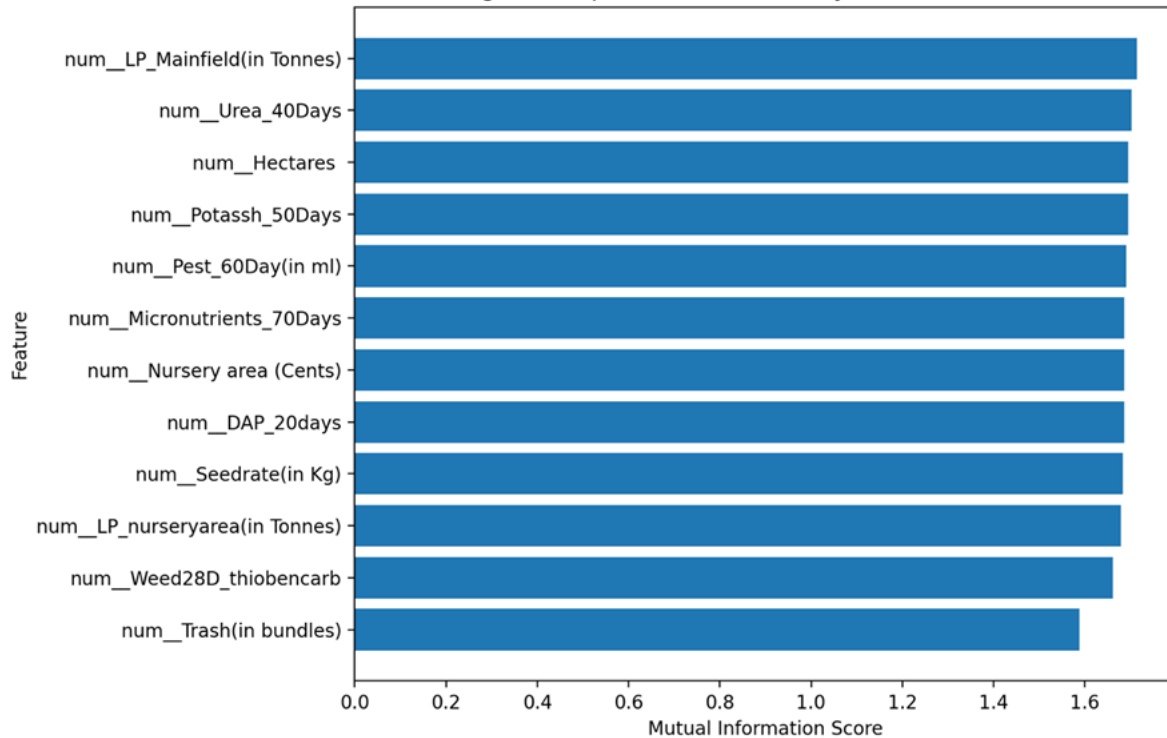


Figure 8. Top Features Selected by Mutual Information

Top predictors identified by the Mutual Information feature selection method. The selected variables are dominated by agricultural input and management-related factors, suggesting that these features contribute most strongly to paddy yield prediction in the dataset.

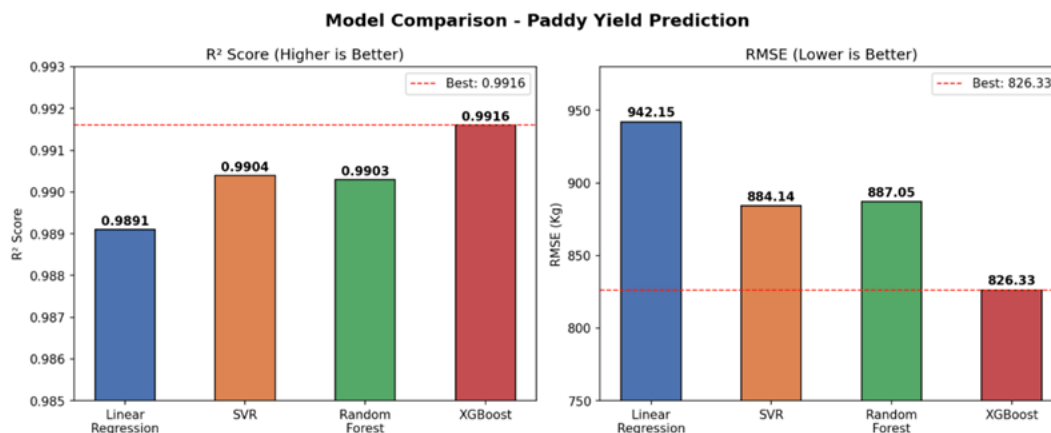


Figure 9: machine learning model comparison among paddy yield

This figure shows moderate agreement between observed and predicted yield categories. The model identified low- and high-yield groups better than the medium-yield group.

Model	RMSE	MAE	R2
Random Forest Regression	208.76	161.89	0.450
Linear Regression baseline	223.35	174.35	0.371

Table 3. Model Performance for Predicting Yield per Hectare

Random Forest Regression performed better than Linear Regression, with lower RMSE and MAE and a higher R². Therefore, Random Forest was used as the main model for creating predicted yield categories.

	Records	Mean yield/ha	Std yield/ha	Variance yield/ha	Min	Median	Max	Productivity level	Stability level
Agriblock									
Sankarapuram	605	6005.32	284.95	81195.70	5410	6050	6469	Higher average productivity	More variable
Kurinjpadi	486	5997.75	284.00	80657.03	5410	5969	6469	Higher average productivity	More stable
Chinnasalem	410	5987.80	285.71	81628.31	5410	6006	6469	Lower average productivity	More variable
Kallakurichi	413	5987.29	280.46	78657.05	5410	5916	6469	Lower average productivity	More stable
Panruti	425	5982.45	284.71	81057.88	5410	5966	6469	Lower average productivity	More variable
Cuddalore	450	5975.95	279.44	78089.24	5410	5916	6469	Lower average productivity	More stable

Table 4. Agriblock-Level Productivity and Stability Summary

This table summarizes productivity and stability by Agriblock. Average yield differences were small, but stability labels helped identify more stable and more variable production zones.

RQ2 Model Performance Summary			
Model	Test R²	Test RMSE	Overfit Gap
Linear Regression	0.9894	927.94	0.0004
Random Forest	0.9903	887.32	0.0037
XGBoost	0.9906	872.44	0.0034
Linear Regression + MI	0.9894	927.64	0.0004
Random Forest + MI	0.9915	830.22	0.0017
XGBoost + Tree	0.9907	870.61	0.0033

Table 5. Comparison of Test R² Across Models

Comparison of predictive performance across baseline and feature-selected models using test-set R². The Random Forest model with Mutual Information feature selection achieved the highest predictive accuracy, indicating that feature selection provided a modest improvement in generalization performance over models trained on all features.